Specification

1 Purpose and scope

This document sets the design assumptions (ASM), requirements (REQ) and selection criteria (CRIT) common to the ALIMOCHE preliminary design. It develops ALM-00 and integrates the outputs of its annexes. It does not choose a configuration or geometry, nor does it declare regulatory compliance.

2 Input data

Table 1 lists the input documents and Table 2 the conventions used.

Table 1. Input documents.

Code	Document
ALM-00	Problem statement and design basis
ALM-P00-A01	Legal study
ALM-P00-A02	Study of similar aircraft
ALM-P00-A03	Materials study
ALM-P00-A04	Operations study
ALM-P00-A05	Powerplant and upgrades study

Table 2. Conventions.

Code	Meaning
ASM	Design assumption: data and calculation rules common to all proposals. Using other values requires a change request.
REQ	Requirement. Mandatory: failing it eliminates the proposal. Target: scored through a CRIT.
CRIT	Selection criterion, with a weight and a scale from 0 to 10.
TBD	Value still to be defined; the annex or person responsible for closing it is stated.

3 Design assumptions

ASM-01 **Flight environment**

Design case. Lillo aerodrome (LELT), elevation ≈ 681 m, 40 °C, ISA pressure at that elevation. Density ≈ 1.04 kg/m³ (σ ≈ 0.85 density altitude ≈ 1700 m).

Comparison case. ISA at sea level.

ASM-02 **Pilot and payload**

Pilot. Equipped (clothing and helmet), from 50 to 90 kg. The design is trimmed with 75 kg and no ballast. The three load cases are given in Table 3.

Table 3. Pilot load cases.

Case	Pilot	Ballast	Use
M1 · Light	50 kg	Up to matching the CG of M2	Aft CG limit
M2 · Design	75 kg	None	Trim and performance
M3 · Maximum	90 kg	None	Structural case; forward CG limit

ASM-03 **Calculation basis**

Reference regulations. To be defined with the legal study (A01), which will set the load and safety factors and the load cases. Candidate: CS-22 as voluntary guidance.

Estimation methods and tools. Common CLmax, Oswald factor, drag and mass-breakdown template: TBD (A02). Hand calculation and a bar-element FEM validated against cases with known solutions.

4 Requirements

Table 4 lists the requirements, their type, origin and method of verification.

Table 4. Requirements.

ID	Requirement	Type	Origin	Verification
REQ-01	Empty mass less than 70 kg. Exact definition TBD (A01).	Mandatory	ALM-00	Mass
REQ-02	Accept pilots from 50 to 90 kg (cases M1–M3).	Mandatory	ALM-00; ASM-02	CG position
REQ-03	Withstand the loads of ASM-03.	Mandatory	ALM-00; ASM-03	Calculation and test
REQ-04	Mission: bungee launch from level ground and straight glide at low height. Height and wind TBD (A04).	Mandatory	ALM-00; A04	Performance
REQ-05	Metal-tube and fabric-covered construction.	Mandatory	Project	Review
REQ-06	Stable and predictable flight, as a basic trainer. Quantitative criteria TBD.	Mandatory	ALM-00	Stability
REQ-07	CG within limits in cases M1, M2 and M3.	Mandatory	ASM-02	CG position
REQ-08	Transportable by road. Envelope TBD (A04).	Mandatory	ALM-00; A04	Dimensions
REQ-09	Buildable in a small workspace with simple means.	Mandatory	ALM-00	Processes
REQ-10	Materials and components available in Spain.	Mandatory	ALM-00; A03	Materials
REQ-11	Materials cost of 3500 ± 500 €.	Target	ALM-00	Budget
REQ-12	Construction in 150 person-hours.	Target	ALM-00	Hours
REQ-13	Easy to inspect, repair and maintain.	Target	ALM-00	Review
REQ-14	Structure suitable for car or winch tow and aerotow.	Target	Project	Calculation

5 Selection criteria

Each proposal that meets the mandatory REQs is scored from 0 to 10 on each criterion according to its scale, and its final score is the weighted sum, with weights fixed before the proposals are seen (Table 5).

Table 5. Selection criteria.

ID	Criterion	Weight	Score 0	Score 10
CRIT-01	Mass margin against 70 kg	25 %	≥ 70 kg	Margin ≥ 15 %
CRIT-02	Simplicity of construction	20 %	External processes	Drilling and riveting
CRIT-03	Transport and assembly	15 %	More than 4 people	2 people without tools
CRIT-04	Trainer flying qualities	15 %	Marginal stability	Stable in 3 axes
CRIT-05	Inspection and repair	10 %	Hidden joints	Visible joints
CRIT-06	Cost and construction hours	10 %	> 4000 € or > 200 h	≤ 3000 € and ≤ 150 h
CRIT-07	Growth potential	5 %	Prevents growth	Meets REQ-14

Open. Weights and scales are proposed; they are to be fixed at a meeting of the preliminary-design core team.